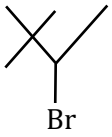
EXERCISE (O-1)

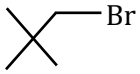
SINGLE CORRECT TYPE QUESTIONS

Elimination reaction (E2) [Base like alc.KOH, $NaNH_2$, RLi, t-BuOK]

1. Which of the following cannot undergo E2 reaction?

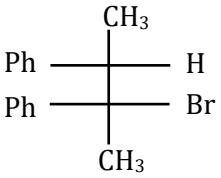
(A) 

(B) 

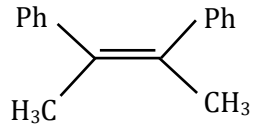
(C) 

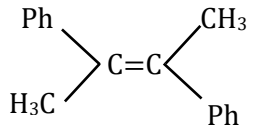
(D) None of these

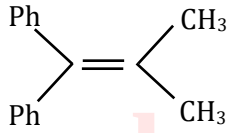
CRM001

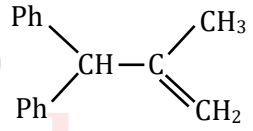
2. 

alc. KOH → Product can be

(A) 

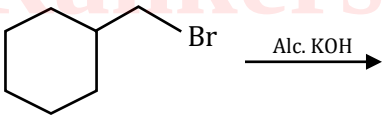
(B) 

(C) 

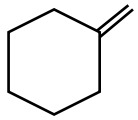
(D) 

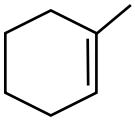
CRM002

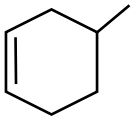
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3. 

The product obtained when it is subjected to E2 reaction will be

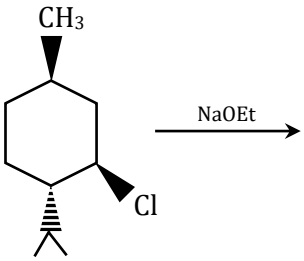
(A) 

(B) 

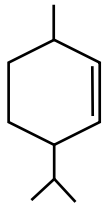
(C) 

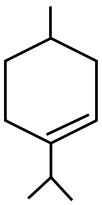
(D) All of these

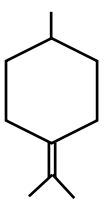
CRM003

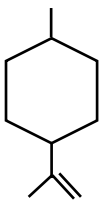
4. 

Major product will be

(A) 

(B) 

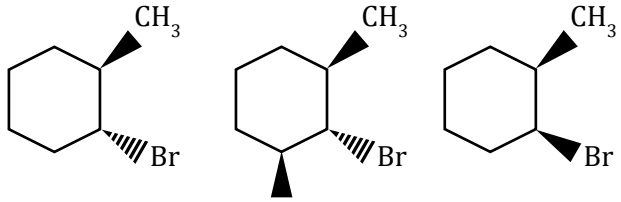
(C) 

(D) 

CRM004

For More Join: @IITJEE_Advanced

5. Arrange the following in order of decreasing rate of E2 reaction

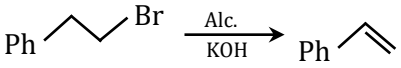


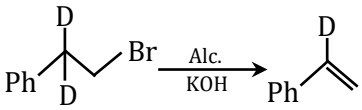
(P) (Q) (R)

(A) P > Q > R (B) R > P > Q (C) R > Q > P (D) Q > P > R

CRM005

6. In the following reactions:

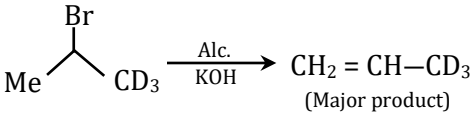
I. 

II. 

The rate of reaction of (I) and (II) are same. Both reactions proceed by which mechanism.

(A) E1 (B) E2 (C) E1 cB (D) Anti-elimination

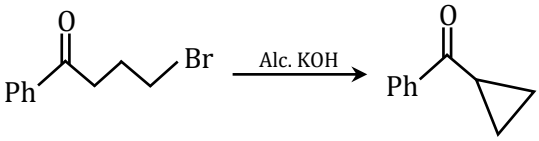
CRM006

7. 

By which mechanism does the above reaction proceed?

(A) E1 (B) E2 (C) E1 cB (D) γ-Elimination

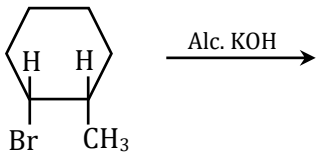
CRM007

8. 

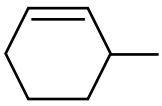
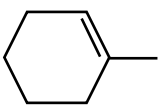
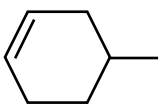
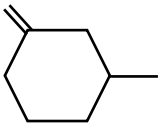
By which mechanism does the above reaction proceed?

(A) α-Elimination (B) β-Elimination (C) γ-Elimination (D) δ-Elimination

CRM008

9. 

The major product is

(A)  (B)  (C)  (D) 

CRM009

Elimination reaction (E1)

10. Which product is formed in E1 reaction of 1–Bromo-2, 3–dimethylcyclohexane?

(A) 3, 4–Dimethylcyclohexane (B) 2, 3–Dimethylcyclohexane

(C) 4, 5–Dimethylcyclohexene (D) 1, 2–Dimethylcylohexene

CRM010

For More Join: @IITJEE_Advanced

11. Which alkyl halide would you expect to be least reactive in E1 reaction ?

(A) $CH_2 = CH - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_2$

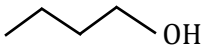
(B) $Ph - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3$

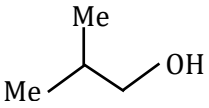
(C) $CH_2 = \underset{\displaystyle Br}{\underset{|}{C}} - CH_2 - CH_3$

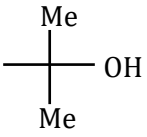
(D) $Ph - CH_2 - CH_2\,Br$

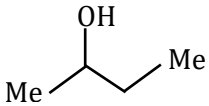
CRM011

12. The compound which would give the most stable carbocation on dehydration is:

(A) 

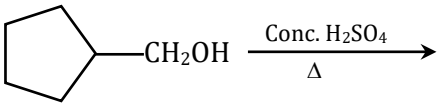
(B) 

(C) 

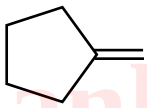
(D) 

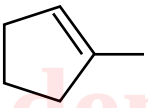
CRM012

13. For the reaction,

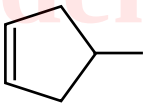


the major product is

(A) 

(B) 

(C) 

(D) 

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CRM013

Nucleophilic Substitution reaction (S_N2) [Nucleophile such as KSH, RSK, $AgNO_2$, KNO_2, AgCN, KCN, $NaNO_2$, Carbanion (R-), NaX, aq.KOH, ROK, RNH_2]

14. Which of the following is not an electrophile?

(A) $H^{\oplus}$ (B) BF_3 (C) $^{\oplus}NO_2$ (D) NH_3

CRM014

15. Which of the following is not a nucleophile?

(A) $FeBr_3$ (B) $Br^{\ominus}$ (C) NH_3 (D) 

CRM015

16. How are basicity and leaving group ability related?

(A) They are not related to each other
(B) Good leaving groups are strong bases
(C) Good leaving groups are weak bases
(D) Leaving group $\propto$ Basic strength

CRM016

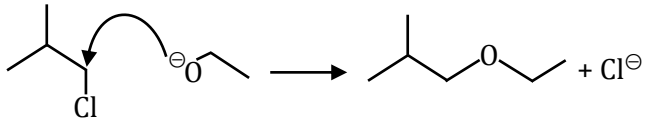
17. Ethers can act as :

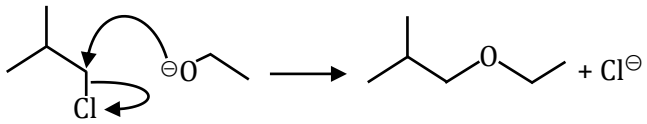
(A) Bronsted acids
(B) Bronsted bases
(C) Lewis acids
(D) Lewis bases

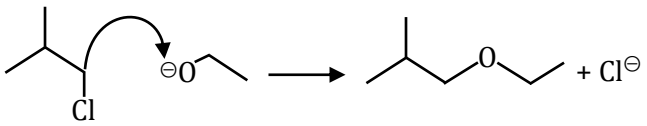
CRM017

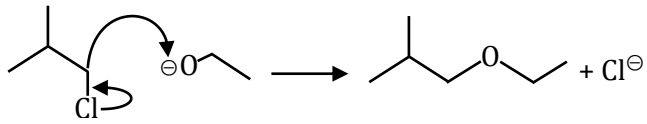
For More Join: @IITJEE_Advanced

18. Which of the following reactions shows the correct use of "curly arrows"?

(A) 

(B) 

(C) 

(D) 

CRM018

19. There are a number of definitions for acids and bases. Match the following definitions to the correct theory.

Theory		Definition	
A	Arrhenius	I	Donates or accepts protons
B	Bronsted-Lowry	II	Donates or accepts a lone pair of electrons
C	Lewis	III	Donates a proton or a hydroxide

(A) A(I), B(II), C(III) (B) A(I), B(III), C(II) (C) A(II), B(III), C(II) (D) A(III), B(I), C(II)

CRM019

20. **Assertion :** Ethers behave as bases in the presence of acids.
Reason : This is due to the presence of lone pairs of electrons on oxygen.
(A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(C) If Assertion is True but the Reason is False.
(D) If both Assertion & Reason are false.

CRM020

21. Identify set of electrophiles :

(A) CO_2, $\overset{\oplus}{C}H_3$, $:CH_2$, Br_2 (B) HOH, SO_3, $:\overset{\ominus}{C}Cl_2$, $Cl^{\oplus}$

(C) SO_2, $CH_3-\ddot{\underset{\ddot{}}{O}}H$, $\overset{\oplus}{N}O_2$, $:\overset{\ominus}{C}(Cl)(F)$ (D) $H-\overset{\oplus}{C}=O, \overset{\oplus}{N}=O, Ph-\overset{\oplus}{C}H_2, \ddot{N}H_3$

CRM021

22. Incorrect statement about carbocation is :
(A) It is lewis acid (B) It has 6 electrons in valency shell
(C) It is electrophile (D) It is always trigonal planar

CRM022

23. Which is the best leaving group?

(A) $H_3C-O^{\ominus}$ (B) $H_3C-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-O^{\ominus}$ (C) $H_3C-CH_2^{\ominus}$ (D) $H_3C-NH^{\ominus}$

CRM023

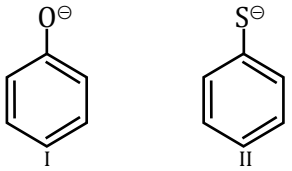
For More Join: @IITJEE_Advanced

24. Which is the MOST nucleophilic in the following series?

(A) $F^{\ominus}$ (B) CH_3CH_2OH (C) H_2O (D) $CH_3CH_2O^{\ominus}$

CRM024

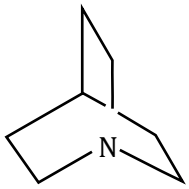
25. Consider the following two anionic molecules. Which of the following statements is TRUE ?

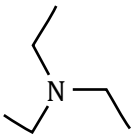


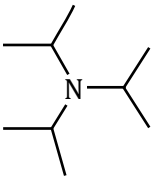
(A) I is more basic and more nucleophilic in PPS than II.
(B) I is less basic and less nucleophilic in PPS than II.
(C) I is more basic but less nucleophilic in PPS than II.
(D) I is less basic but more nucleophilic in PPS than II.

CRM025

26. Identify correct nucleophilicity order

(i) 

(ii) 

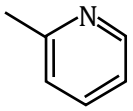
(iii) 

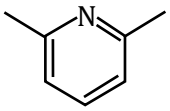
(A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

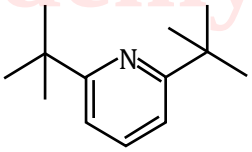
CRM026

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27. Identify correct nucleophilicity order

(i) 

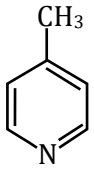
(ii) 

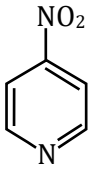
(iii) 

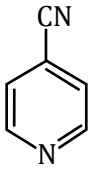
(A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

CRM027

28. Identify correct nucleophilicity order

(i) 

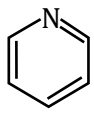
(ii) 

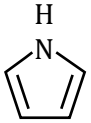
(iii) 

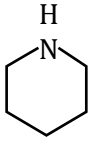
(A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

CRM028

29. Identify correct nucleophilicity order

(i) 

(ii) 

(iii) 

(A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

CRM029

For More Join: @IITJEE_Advanced

30. Identify correct nucleophilicity order

(i) H_2O (ii) H_2S (iii) H_2Se

(A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (i) > (iii) > (ii)

CRM030

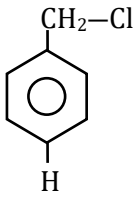
31. Identify correct nucleophilicity order

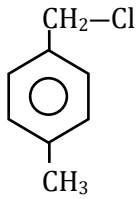
(i) PH_3 (ii) NH_3 (iii) AsH_3

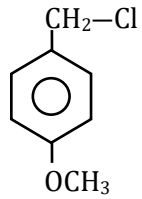
(A) (i) > (ii) > (iii) (B) (ii) > (iii) > (i) (C) (i) > (iii) > (ii) (D) (ii) > (iii) > (i)

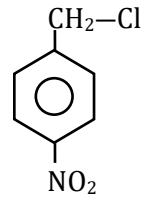
CRM031

32. Which of the following is most reactive toward S_N2.

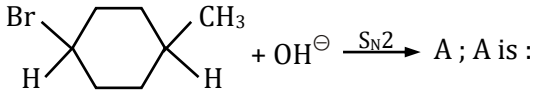
(A) 

(B) 

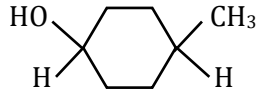
(C) 

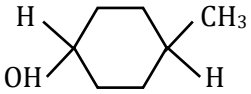
(D) 

CRM032

33. 

A ; A is :

(A) 

(B) 

(C) Both (D) None

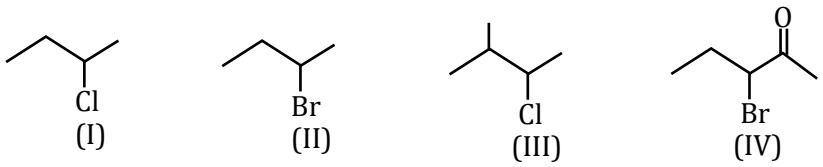
CRM033

34. The reactivity of 2-bromo-2-methylbutane (I), 1-bromopentane (II) and 2-bromopentane (III) towards S_N2 displacement is such that :

(A) I > II > III (B) I > III > II (C) II > III > I (D) II > I > III

CRM034

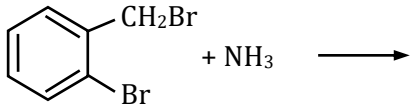
35. Arrange these compounds in order of increasing S_N2 reaction rate :

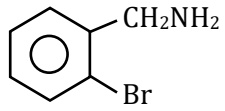


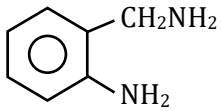
(A) III < I < II < IV (B) III < II < I < IV (C) IV < III < I < II (D) III < IV < I < II

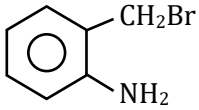
CRM035

36. The major product in the given reaction:



(A) 

(B) 

(C) 

(D) All of these

CRM036

For More Join: @IITJEE_Advanced

37. When ethyl bromide is treated with moist Ag_2O, the main product is:

(A) Ethyl ether (B) Ethanol

(C) Ethoxy ethane (D) All of these

CRM037

38. Which of the following is most reactive for S_N2 reactions.

(A) $C_6H_5(CH_3)(C_6H_5)CBr$ (B) $C_6H_5CH(CH_3)Br$

(C) $(C_6H_5)_3CBr$ (D) $C_6H_5CH_2Br$

CRM038

39. Select suitable reason for non-occurrence of the following reaction ?

$$Br^{\ominus} + CH_3OH \longrightarrow CH_3Br + OH^{\ominus}$$

(A) attacking nucleophile is stronger one

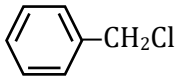
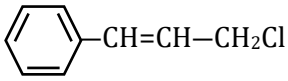
(B) leaving group is strong base than nucleophile

(C) Alcohols are not good substrate for S_N reaction.

(D) Hydroxide ions are weak bases

CRM039

40. Arrange the following in the order of their reactivity of S_N2 reaction

(I)  (II) 

(III) $CH_2=CH-CH_2Cl$ (IV) $CH_3CH_2CH_2Cl$

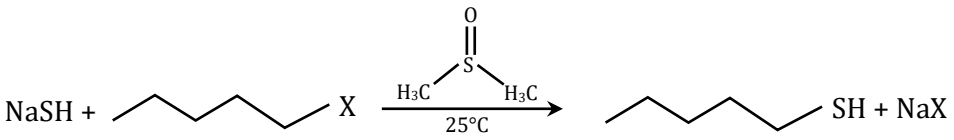
(A) (IV) > (III) > (I) > (II) (B) (II) > (I) > (III) > (IV)

(C) (I) > (II) > (IV) > (III) (D) (III) > (II) > (I) > (IV)

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CRM040

41. For which leaving group X would you predict the S_N2 reaction shown below to take place most rapidly?



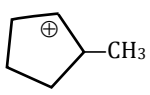
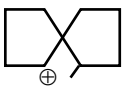
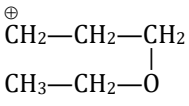
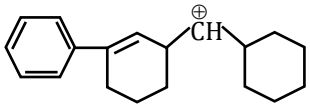
(A) X =–OH (B) X =–Cl (C) X= –Br (D) X=–I

CRM041

Nucleophilic Substitution reaction (S_N1)

42. How many following carbocation undergo re-arrangement –

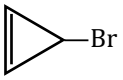
(a) $CH_3CH_2\overset{\oplus}{C}H_2$ (b) $(CH_3)_2\overset{\oplus}{C}HCHCH_3$ (c) $(CH_3)_3\overset{\oplus}{C}CHCH_3$ (d) $(CH_3CH_2)_3\overset{\oplus}{C}CH_2$

(e)  (f)  (g)  (h) 

(A) 5 (B) 8 (C) 6 (D) 7

CRM042

For More Join: @IITJEE_Advanced

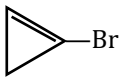
43.  ,which is not the correct statement:

(I)

(A) I is more soluble in water than bromocyclopropane

(B) I gives pale yellow ppt. on addition with aq. $AgNO_3$

(C) I is having lower dipole moment than bromocyclopropane

(D) I is more ionic than 

(I)

CRM043

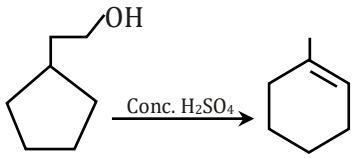
44. A solution of (–) –1–chloro–1–phenylethane in toluene racemises in the presence of a small amount of $SbCl_5$, due to the formation of :-

(A) carbanion (B) Carbene

(C) Carbocation (D) free radical

CRM044

45. How many 1,2-Shifts of carbocation intermediate are involved during the course of following reaction :

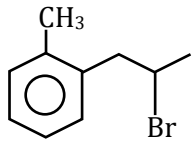


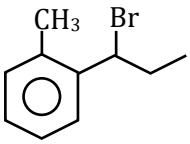
(A) 1 (B) 2 (C) 3 (D) 4

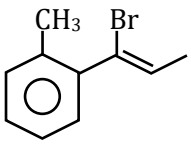
CRM045

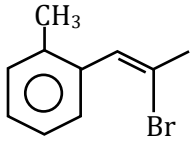
Rankers Academy JEE

46. Which compound undergoes hydrolysis by the S_N1 mechanism at the fastest rate?

(A) 

(B) 

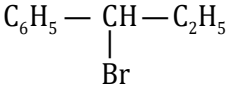
(C) 

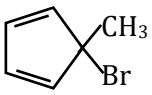
(D) 

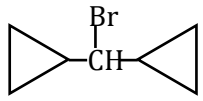
CRM046

47. Arrange the following compounds in decreasing order of their reactivity for hydrolysis reaction

(I) $C_6H_5—CH_2—Br$

(II) 

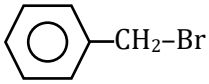
(III) 

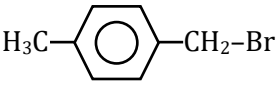
(IV) 

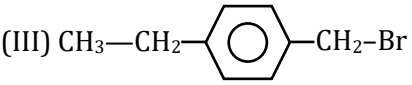
(A) II > III > IV > I (B) IV > II > I > III (C) III > IV > II > I (D) I > II > III > IV

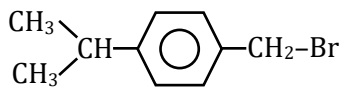
CRM047

48. Arrange the following compounds in order of decreasing rate of hydrolysis for S_N1 reaction:

(I) 

(II) 

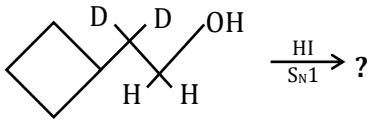
(III) 

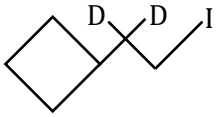
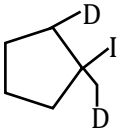
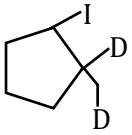
(IV) 

(A) II > III > IV > I (B) IV > III > II > I (C) III > IV > II > I (D) I > II > III > I

CRM048

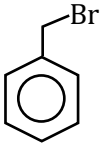
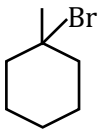
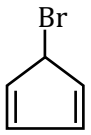
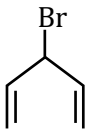
For More Join: @IITJEE_Advanced

49.  ?

(A) 
(B) 
(C) 
(D) None of these

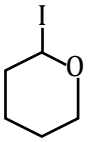
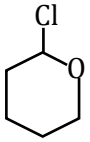
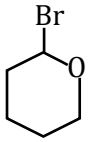
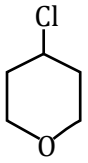
CRM049

50. Which of the following can not give S_N1 reaction easily?

(A) 
(B) 
(C) 
(D) 

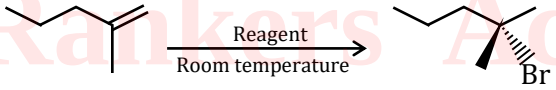
CRM050

51. Which one of the following compounds will be most reactive for S_N1 reactions?

(A) 
(B) 
(C) 
(D) 

CRM051

52. What is the reagent(s) needed for the following reaction?



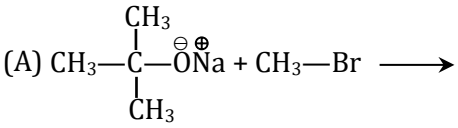
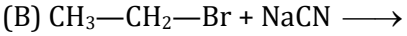
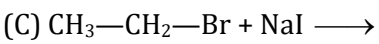
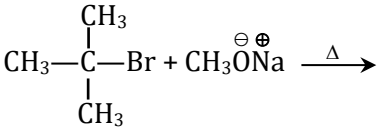
(A) HCl (B) NaBr (C) HBr (D) Br_2/CCl_4

CRM052

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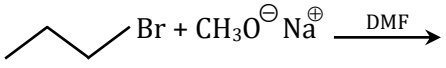
Nucleophilic Substitution reaction vs Elimination reaction (SN vs E)

53. Which reaction gives elimination as a major product

(A) 
(B) 
(C) 
(D) 

CRM053

54. Which reaction is faster (I) or (II) and what is its mechanism?

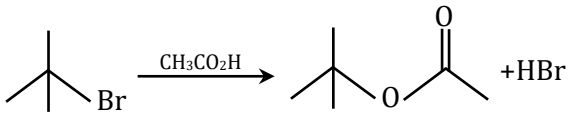
(I) 
(II) 

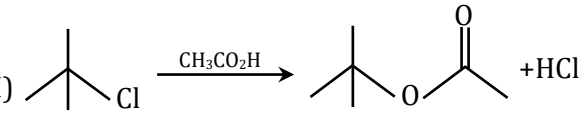
(A) reaction (I), S_N1
(B) reaction (II), S_N2
(C) reaction (II), S_N1
(D) reaction (I), S_N2

CRM054

For More Join: @IITJEE_Advanced

55. Which reaction is faster (I) or (II) and what is its mechanism?

(I) 

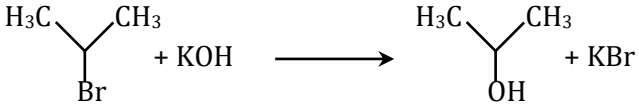
(II) 

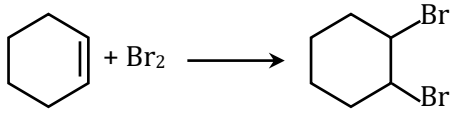
(A) reaction (II), S_N1 (B) reaction (II), S_N2 (C) reaction (I), S_N1 (D) reaction (I), S_N2

CRM055

56. For the following reactions :-

(a) $CH_3CH_2CH_2Br + KOH \longrightarrow CH_3CH{=}CH_2 + KBr + H_2O$

(b) 

(c) 

Which of the following statement is correct?

(A) (a) and (b) are elimination reaction and (c) is addition reaction

(B) (a) is elimination, (b) is substitution and (c) is addition reaction

(C) (a) is elimination, (b) and (c) are substitution reactions

(D) (a) is substitution, (b) and (c) are addition reaction

CRM056

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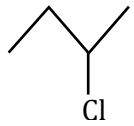
For More Join: @IITJEE_Advanced

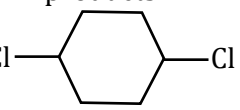
EXERCISE (O-2)

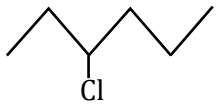
MULTIPLE CORRECT TYPE QUESTIONS

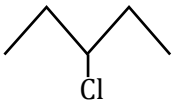
Elimination reaction (E2) [Base like alc.KOH, $NaNH_2$, RLi, t-BuOK]

1. Which of the following is expected to give more than 2 products in E2 reaction?

(A) 

(B) 

(C) 

(D) 

CRM057

Elimination reaction (E1)

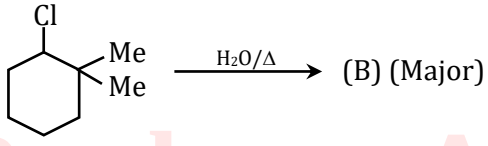
2. Which of the following statement correctly describe(s) E1 reactions of alkyl halides (RX)?

(A) Rate = k[base] [RX] (B) Rate = k [RX]

(C) The reactions occur in two distinct steps (D) Rearrangments are sometimes seen

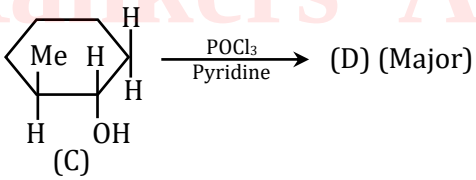
CRM058

3. Consider the following reaction:

I. 

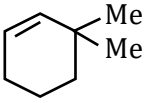
(A)

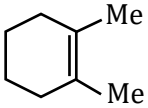
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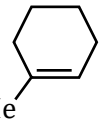
II. 

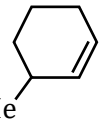
(C)

Which of the following statements are correct?

(A) (B) is 

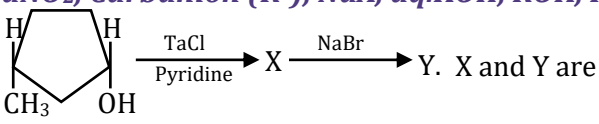
(B) (B) is 

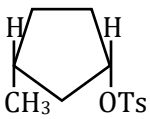
(C) (D) is 

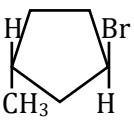
(D) (D) is 

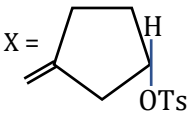
CRM059

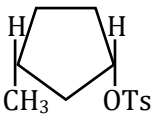
Nucleophilic Substitution reaction (S_N2) [Nucleophile such as KSH, RSK, $AgNO_2$, KNO_2, AgCN, KCN, $NaNO_2$, Carbanion (R-), NaX, aq.KOH, ROK, RNH_2]

4. 

(A) X = 

(B) Y = 

(C) 

(D) Y = 

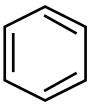
CRM060

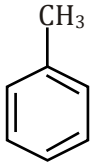
For More Join: @IITJEE_Advanced

5. Which of following is/are not a set of nucleophiles?

(A) $\overset{\ominus}{Cl}, \overset{\ominus}{O}H, \overset{\ominus}{R}, BCl_3$

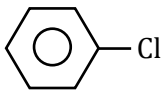
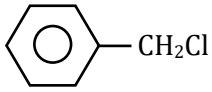
(B) $CH_3-\overset{O}{\overset{\|}{C}}-O^{\ominus}, \overset{\ominus}{N}_3, H_3O^{\oplus}, \overset{\ominus}{S}H$

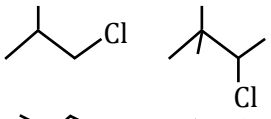
(C) $\ddot{N}H_3$, $CH_3-\ddot{N}H_2$, $CH_3-CH_2-\ddot{O}H$, 

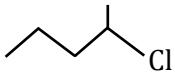
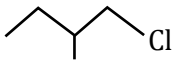
(D) $H^{\ominus}$, $:\overset{\ominus}{C}N$, CS_2, 

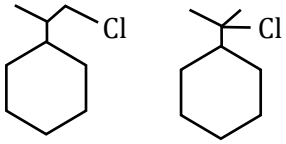
CRM061

6. In the given pair in which pair the first compound is more reactive than second towards S_N2 reaction.

(A)  

(B) 

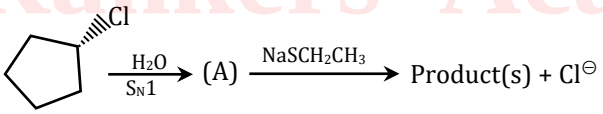
(C)  

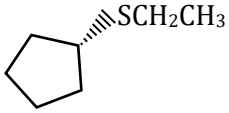
(D) 

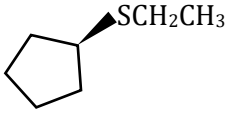
CRM062

Nucleophilic Substitution reaction (S_N1)

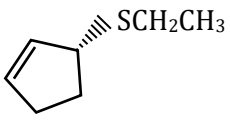
7. What could be the product(s) for the following S_N1 reaction?



(A) 

(B) 

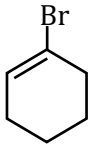
(C) 

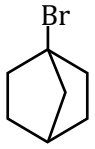
(D) 

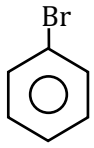
CRM063

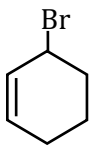
Nucleophilic Substitution reaction vs Elimination reaction (SN vs E)

8. S_N2 reaction will be negligible in

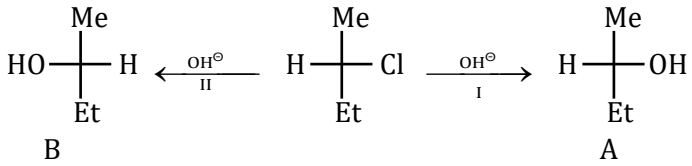
(A) 

(B) 

(C) 

(D) 

CRM064

9. 

Mechanism involved is/are:

(A) I can't be S_N1 (B) II can't be S_N2 (C) I is S_N1 (D) II is S_N2

CRM065

For More Join: @IITJEE_Advanced

10. Incorrect statement among the following is/are :

(A) R–OH with NaI in the presence of phosphoric acid gives R—I, but not in the absence of phosphoric acid

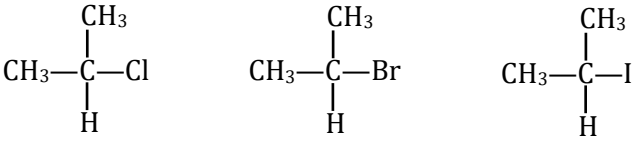
(B) 2-methyl propane on chlorination (Cl_2, hv) gives 1-chloro-2-methyl propane while bromination (Br_2, hv) gives 2-bromo-2-methyl propane

(C) Usually higher temperature prefers substitution over elimination

(D) Triphenyl chloromethane cannot be hydrolysed

CRM066

11. From left to right, correct statements are :



(A) Rate of S_N1 mechanism increases in polar protic solvent

(B) Rate of S_N2 mechanism increases in DMSO

(C) Rate of E_2 mechanism increases

(D) Rate of E_1 mechanism increases

CRM067

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For More Join: @IITJEE_Advanced

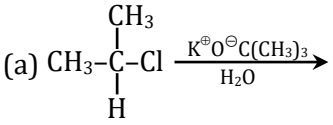
EXERCISE (S)

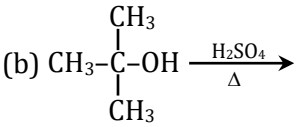
Part-1 : JEE Main (Numerical Value type)

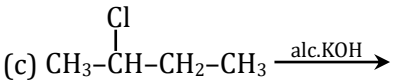
- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

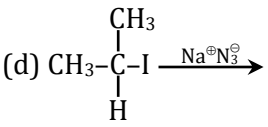
Elimination reaction (E2) [Base like alc.KOH, $NaNH_2$, RLi, t-BuOK]

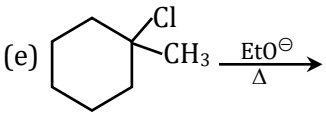
1. How many of the following reagent combinations will result in elimination reactions leading to the major product?

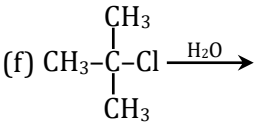




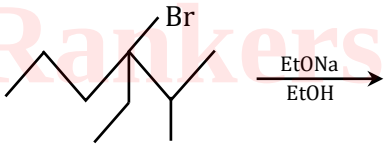








CRM068

2. 

How many total alkenes can be made?

CRM069

3. 2–Bromopentane + $EtO^{\ominus} \xrightarrow{\Delta}$ The total products are

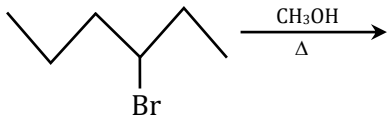
CRM070

Elimination reaction (E1)

4. How many major products can be formed by E1 reaction of 1–chloro–2, 2–diphenyl propane ?

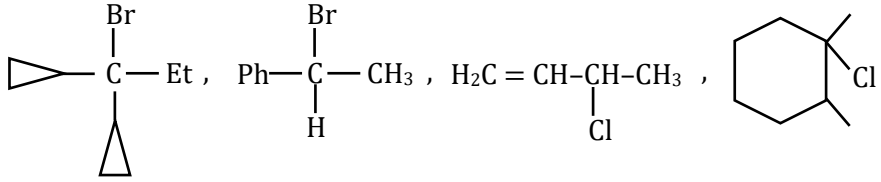
CRM071

5. Find number of possible E1 products from following reaction :



CRM072

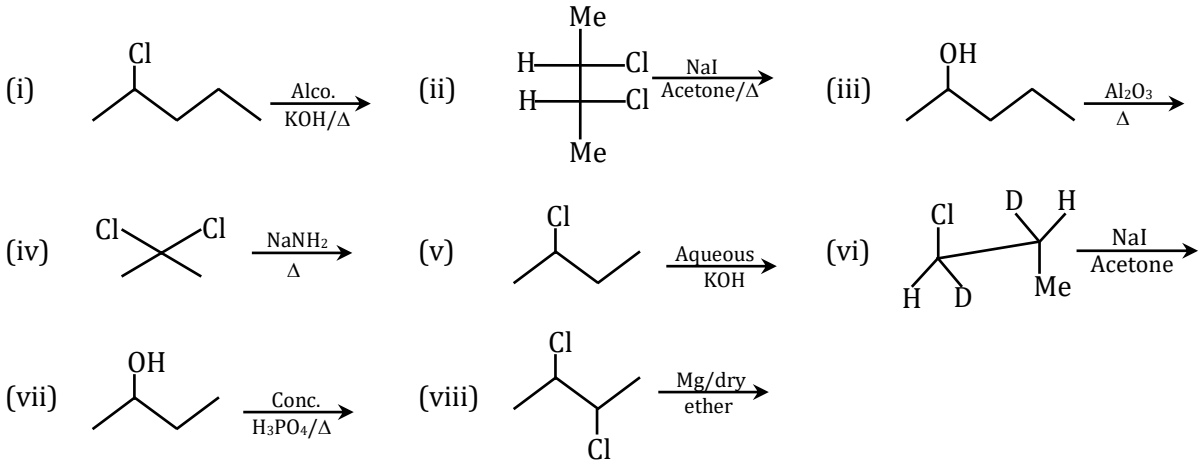
6. How many of the following compounds will give E1 reaction?



CRM073

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For More Join: @IITJEE_Advanced

7. Identify total number of 'β'–elimination reactions



CRM074

8. Calculate total number of alkene products when 2–chloro–2–cyclobutyl hexane react with alcoholic KOH and heat.

CRM075

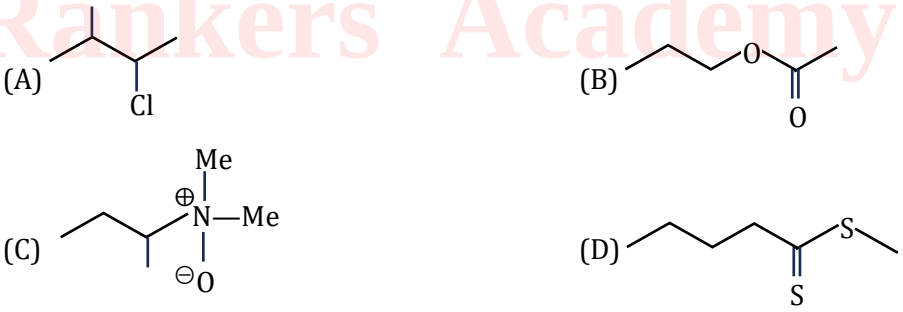
9. Calculate total number of α–H present in major product alkene formed when 2,3–dimethyl butanol react with concentrated H_2SO_4/Δ.

CRM076

Elimination reaction (Ei)[Ester, Xanthates, Amino oxides, Sulfoxide, Selenoxide]

10. Number of following compounds will give syn elimination reaction.

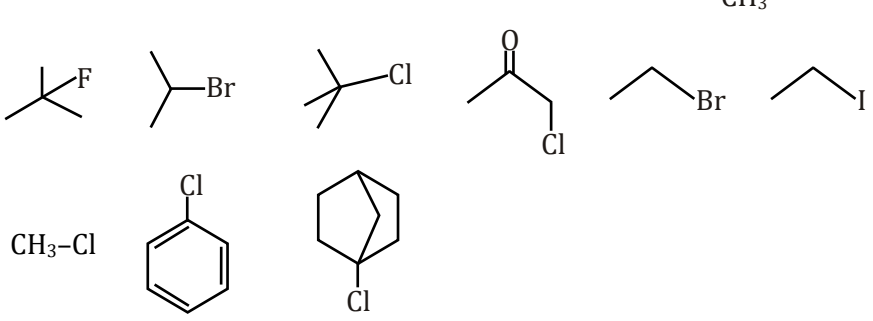
Rankers Academy JEE



CRM077

Nucleophilic Substitution reaction (S_N2) [Nucleophile such as KSH, RSK, $AgNO_2$, KNO_2, AgCN, KCN, $NaNO_2$, Carbanion (R-), NaX, aq.KOH, ROK, RNH_2]

11. Number of compounds which are more reactive than $H_3C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-Cl$ towards S_N2 reaction:



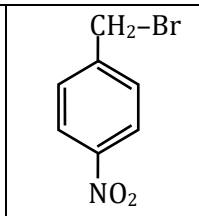
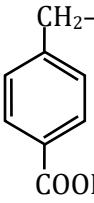
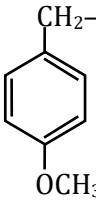
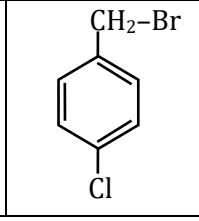
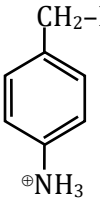
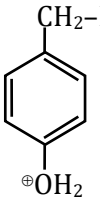
CRM078

For More Join: @IITJEE_Advanced

12. If 1,1–dichloro propane is reacted with aqueous $NaNO_2$ solution, only one of the chlorine undergo nucleophilic substitution (S_N2) reaction. How many mono–substitution products would be formed ? (including stereomers)

CRM079

13. How many of the following compounds encourages "S_N2" mechanism ?

(A)		(B)		(C)	
(D)		(E)		(F)	

CRM080

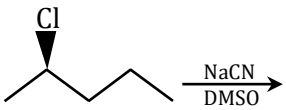
Part-2 : JEE Advanced (Decimal type)

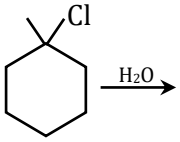
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

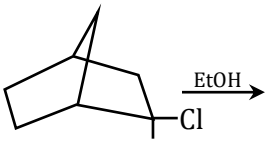
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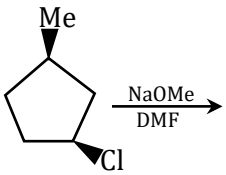
Nucleophilic Substitution reaction (S_N1)

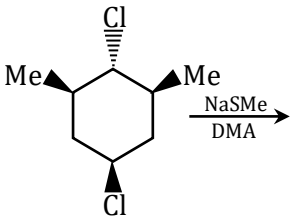
14. Identify total number of S_N1 reactions

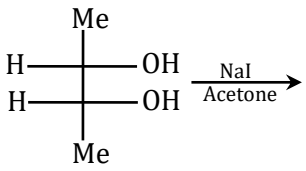
(i) 

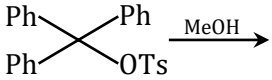
(ii) 

(iii) 

(iv) 

(v) 

(vi) 

(vii) 

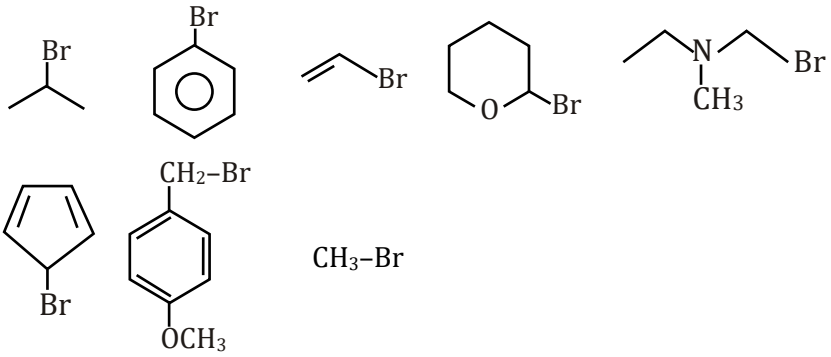
CRM081

15. Calculate percnetage of S_N1 product if (R)–2–chloro butane on reaction with $NaOH/H_2O$ and acetone gives 98 per cent inverted product.

CRM082

For More Join: @IITJEE_Advanced

16. Consider following compounds:



No. of compounds which are more reactive than  for S_N1 reaction are :

CRM083

17. Number of the correct statement among the following

(i) S_N1 reactions involve two step

(ii) S_N2 reactions involve single step

(iii) S_N1 reactions involves carbocation intermediate

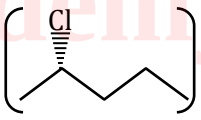
(iv) S_N2 reaction involves carbonium ion intermediate

CRM084

Nucleophilic Substitution reaction vs Elimination reaction (SN vs E)

18. When (S)–2–chloropentane reacts with NaOH in 75 percent EtOH and 25 percent acetone follows rate law :

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Rate = 2.35×10^{-5}  .[OH] + 5.30×10^{-6} 

Then, calculate percentage of S_N1 product when concentration of [OH] = 1.5 molar.

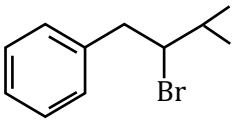
If percentage of S_N1 product = X, then calculate value of Y if.

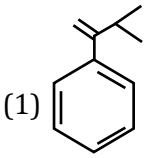
Y = X–5.(nearest integer)

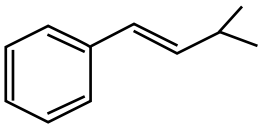
CRM085

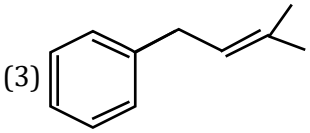
For More Join: @IITJEE_Advanced

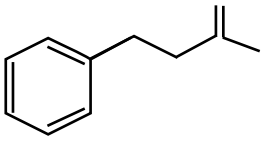
EXERCISE - JEE (Main) PYQ

1.  Major product "P" Product P is **[JEE (Main) 2024]**

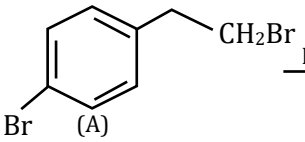
(1) 

(2) 

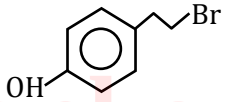
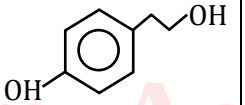
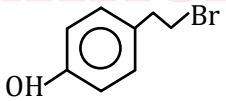
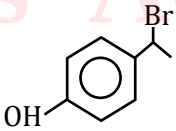
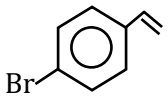
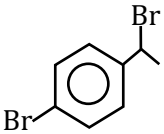
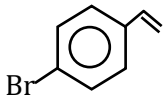
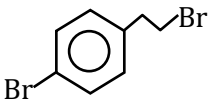
(3) 

(4) 

CRM086

2.  Identify Ⓑ and Ⓒ and how are Ⓐ and Ⓒ related ?

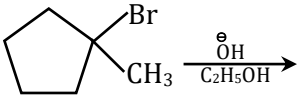
[JEE (Main) 2024]

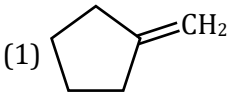
(1)			functional group isomers
(2)			Derivative
(3)			position isomers
(4)			chain isomers

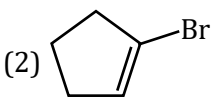
Rankers Academy JEE

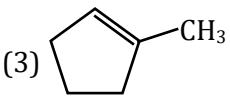
CRM087

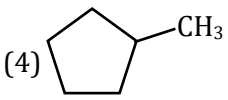
3. Identify the major product in the following reaction. **[JEE (Main) 2024]**



(1) 

(2) 

(3) 

(4) 

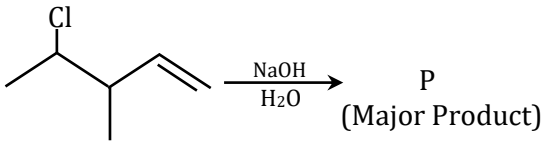
CRM088

For More Join: @IITJEE_Advanced

4. The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement) **[JEE (Main) 2023]**

(1) 1-Bromo-2-methylbutane
(2) 2-Bromopropane
(3) 2-Bromopentane
(4) 2-Bromo-3,3-dimethylpentane

CRM089

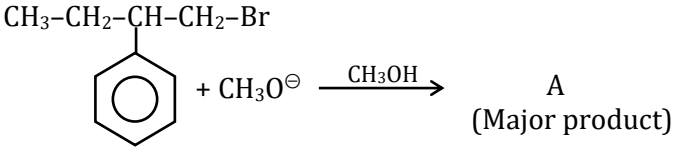
5. 

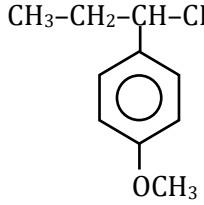
[JEE (Main) 2022]

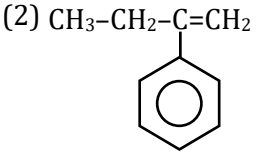
Consider the above reaction. The number of π electrons present in the product 'P' is___.

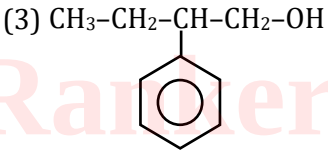
CRM090

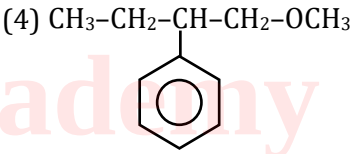
6. The major product (A) formed in the reaction given below is : **[JEE (Main) 2021]**



(1) 

(2) 

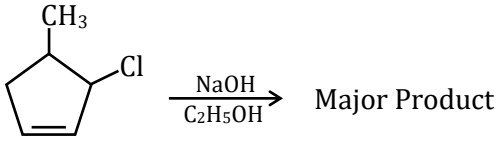
(3) 

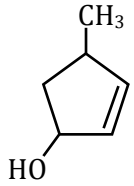
(4) 

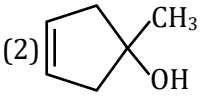
Randers Academy JEE

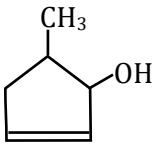
CRM091

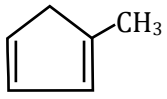
7. The major product of the following reaction is : **[JEE (Main) 2021]**



(1) 

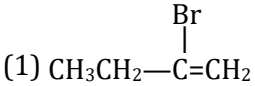
(2) 

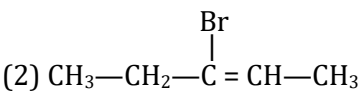
(3) 

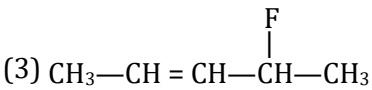
(4) 

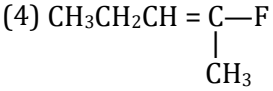
CRM092

8. The major product obtained from E_2-elimination of 3-bromo-2-fluoropentane is: **[JEE (Main) 2020]**

(1) 

(2) 

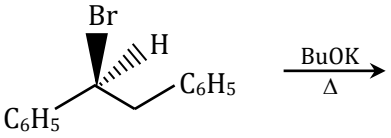
(3) 

(4) 

CRM093

For More Join: @IITJEE_Advanced

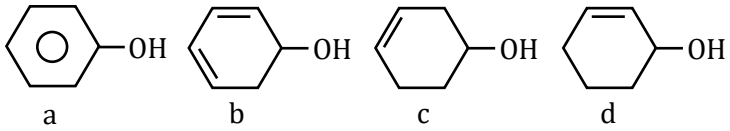
9. The major product obtained in the following reaction is :- **[JEE (Main) 2017]**



(1) $(\pm)C_6H_5CH(O^tBu)CH_2CH_6H_5$
(2) $C_6H_5CH{=}CHC_6H_5$
(3) $(+)C_6H_5CH(O^tBu)CH_2H_5$
(4) $(-)C_6H_5CH(O^tBu)CH_2C_6H_5$

CRM094

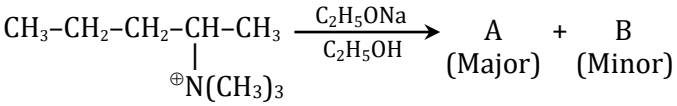
10. Decreasing order of dehydration of the following alcohols is **[JEE (Main) 2023]**



(1) a > d > b > c
(2) b > d > c > a
(3) b > a > d > c
(4) d > b > c > a

CRM095

11. Identify the correct statement for the below given transformation. **[JEE (Main) 2022]**

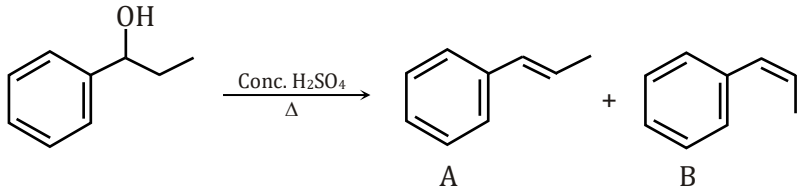


(1) A - $CH_3CH_2CH = CH{-}CH_3$, B – $CH_3CH_2CH_2CH = CH_2$, Saytzeff products
(2) A - $CH_3CH_2CH = CH{-}CH_3$, B – $CH_3CH_2CH_2CH = CH_2$, Hofmann products
(3) A - $CH_3CH_2CH_2CH = CH_2$, B – $CH_3CH_2CH = CHCH_3$, Hofmann products
(4) A - $CH_3CH_2CH_2CH = CH_2$, B – $CH_3CH_2CH = CHCH_3$, Saytzeff products

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CRM096

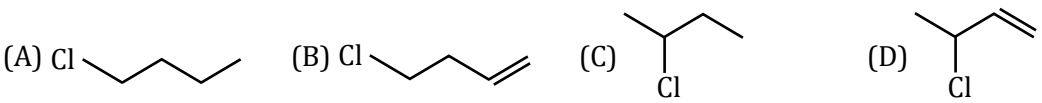
12.



Consider the above reaction, and choose the correct statement : **[JEE (Main) 2021]**
(1) The reaction is not possible in acidic medium
(2) Both compounds **A** and **B** are formed equally
(3) Compound **A** will be the major product
(4) Compound **B** will be the major product

CRM097

13. The decreasing order of reactivity towards dehydrohalogenation (E_1) reaction of the following compounds is : **[JEE (Main) 2020]**

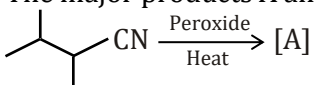


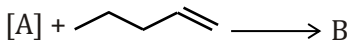
(1) B > D > A > C
(2) B > D > C > A
(3) D > B > C > A
(4) B > A > D > C

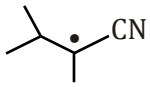
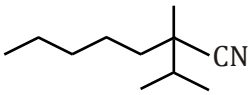
CRM098

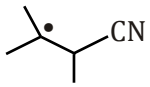
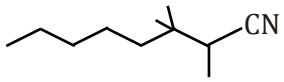
For More Join: @IITJEE_Advanced

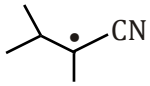
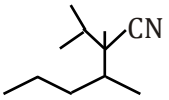
14. The major products A and B in the following reactions are **[JEE (Main) 2020]**

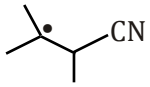
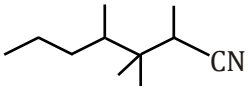




(1) A =  and B = 

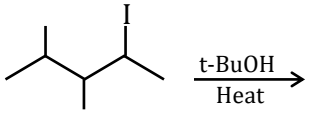
(2) A =  and B = 

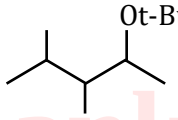
(3) A =  and B = 

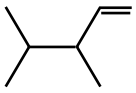
(4) A =  and B = 

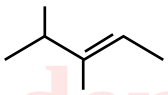
CRM099

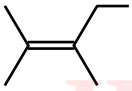
15. The major product in the following reaction is : **[JEE (Main) 2020]**



(1) 

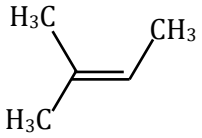
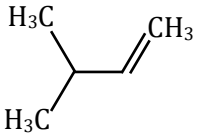
(2) 

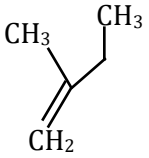
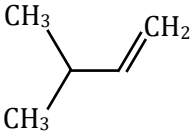
(3) 

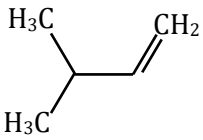
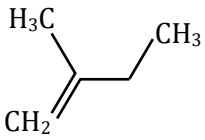
(4) 

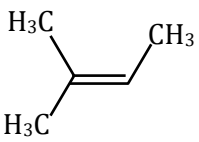
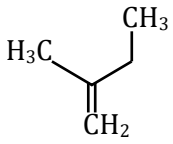
CRM100

16. When neopentyl alcohol is heated with an acid, it slowly converted into an 85 : 15 mixture of alkenes A and B, respectively. What are these alkenes ? **[JEE (Main) 2020]**

(1)  and 

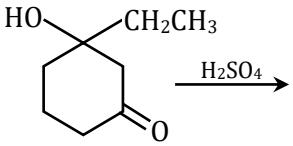
(2)  and 

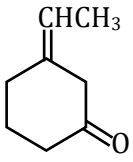
(3)  and 

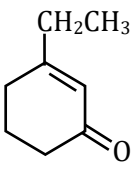
(4)  and 

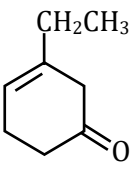
CRM101

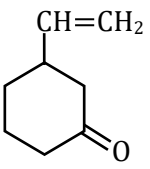
17. The major product of the following reaction is: **[JEE (Main) 2020]**



(1) 

(2) 

(3) 

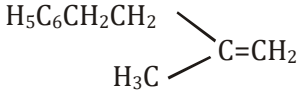
(4) 

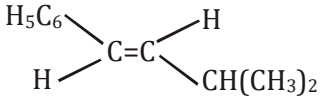
CRM102

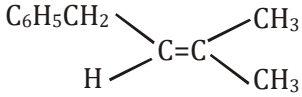
Rankers Academy JEE
For More Join: @IITJEE_Advanced

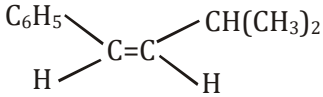
18. The main product of the following reaction is **[AIEEE-2010]**

$C_6H_5CH_2CH(OH)CH(CH_3)_2 \xrightarrow{conc.H_2SO_4} ?$

(1) 

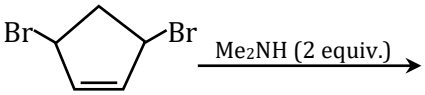
(2) 

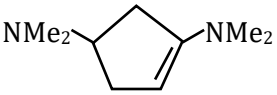
(3) 

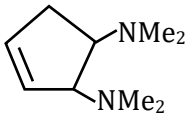
(4) 

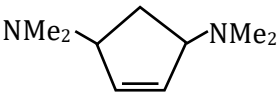
CRM103

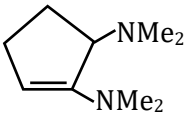
19. Find out the major product formed from the following reaction. [Me: $-CH_3$] **[JEE (Main) 2024]**



(1) 

(2) 

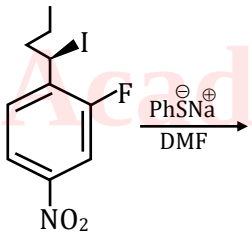
(3) 

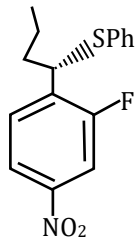
(4) 

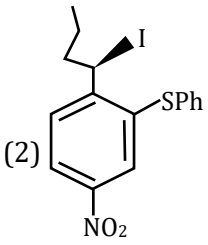
CRM104

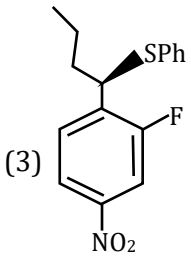
20. The major product of the following reaction is: **[JEE (Main) 2022]**

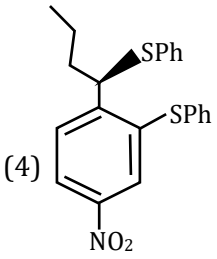
Rankers Academy JEE



(1) 

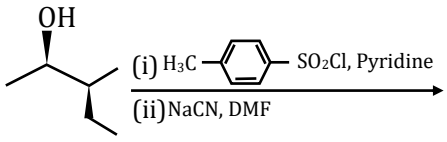
(2) 

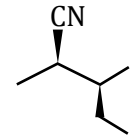
(3) 

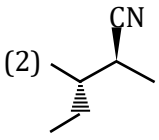
(4) 

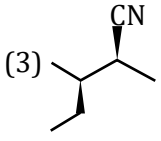
CRM105

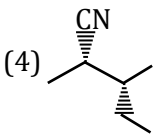
21. Most stable product of the following reaction is: **[JEE (Main) 2022]**



(1) 

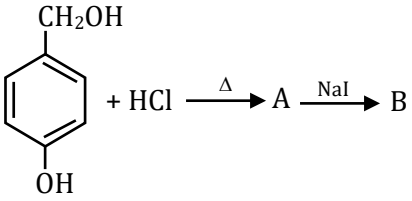
(2) 

(3) 

(4) 

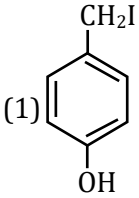
CRM106

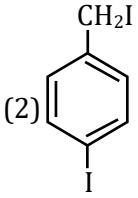
For More Join: @IITJEE_Advanced

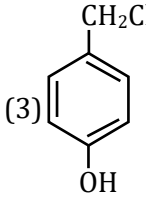
22. 

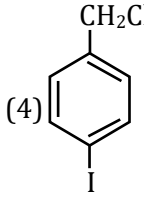
[JEE (Main) 2022]

In the above reaction product B is:

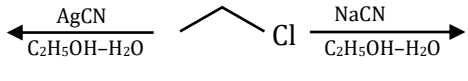
(1) 

(2) 

(3) 

(4) 

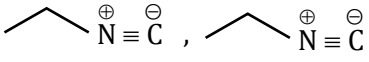
CRM107

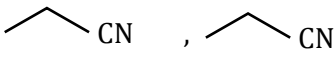
23. 'A' (Major Product) 

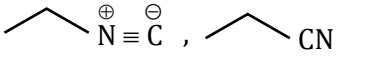
'B' (Major Product)

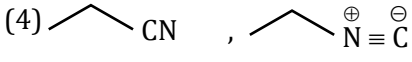
Considering the above reactions, the compound 'A' and compound 'B' respectively are :

[JEE (Main) 2022]

(1) 

(2) 

(3) 

(4) 

CRM108

24. The correct pair(s) of the ambident nucleophiles is (are) : [JEE (Main) 2021]

(A) AgCN/KCN
(B) RCOOAg/RCOOK
(C) $AgNO_2/KNO_2$
(D) AgI/KI

(1) (B) and (C) only (2) (A) only (3) (A) and (C) only (4) (B) only

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CRM109

25. Given below are two statements : [JEE (Main) 2021]

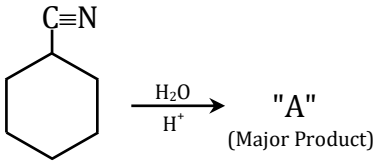
Statement I : C_2H_5OH and AgCN both can generate nucleophile.

Statement II : KCN and AgCN both will generate nitrile nucleophile with all reaction conditions.

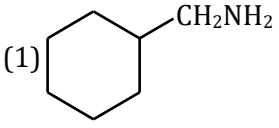
Choose the most appropriate option :

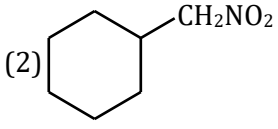
(1) Statement I is true but statement II is false
(2) Both statement I and statement II are true
(3) Statement I is false but statement II is true
(4) Both statement I and statement II are false

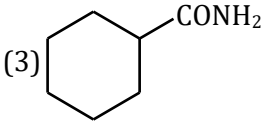
CRM110

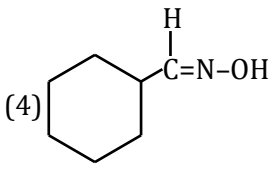
26. 

Consider the above chemical reaction and identify product "A" [JEE (Main) 2021]

(1) 

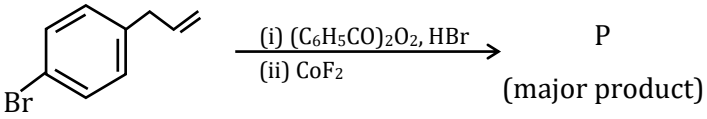
(2) 

(3) 

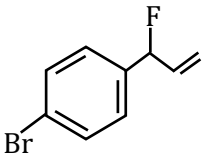
(4) 

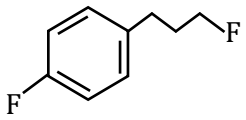
CRM111

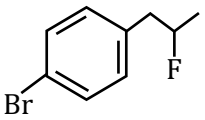
For More Join: @IITJEE_Advanced

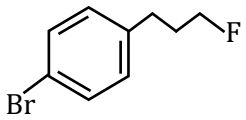
27. 

Major product P of above reaction, is : **[JEE (Main) 2021]**

(1) 

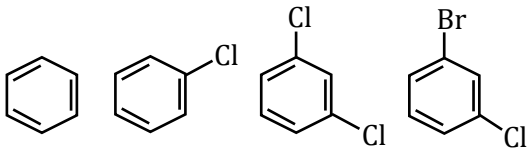
(2) 

(3) 

(4) 

CRM112

28. The correct decreasing order of densities of the following compounds is : **[JEE (Main) 2021]**

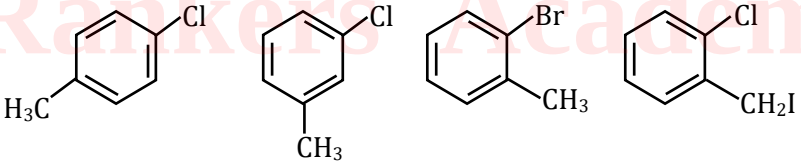


(A) (B) (C) (D)

(1) (D) > (C) > (B) > (A)
(2) (C) > (D) > (A) > (B)
(3) (C) > (B) > (A) > (D)
(4) (A) > (B) > (C) > (D)

CRM113

29. Among the following compounds I-IV, which one forms a yellow precipitate on reacting sequentially with (i) NaOH (ii) dil. HNO_3 (iii) $AgNO_3$? **[JEE (Main) 2021]**

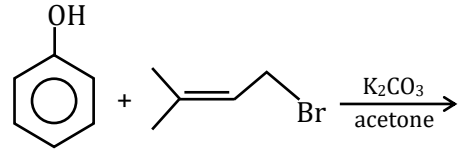


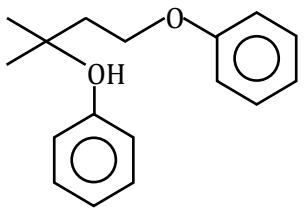
I II III IV

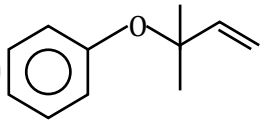
(1) II (2) IV (3) I (4) III

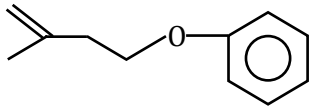
CRM114

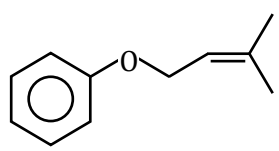
30. The major product of the following reaction, if it occurs by S_N2 mechanism is : **[JEE (Main) 2021]**



(1) 

(2) 

(3) 

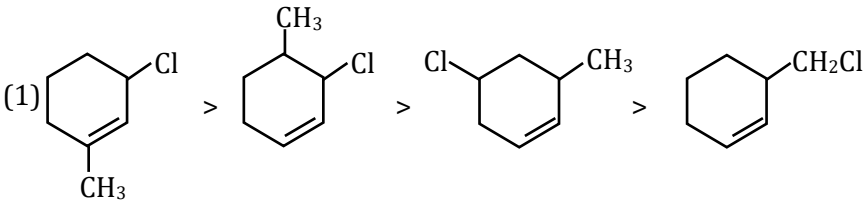
(4) 

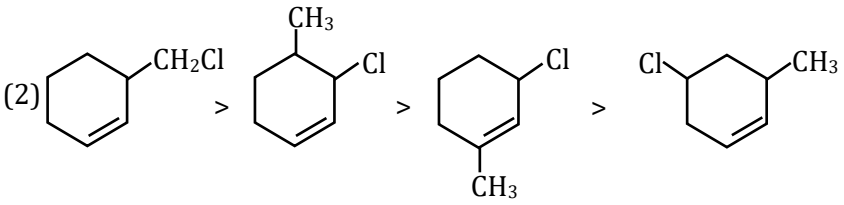
CRM115

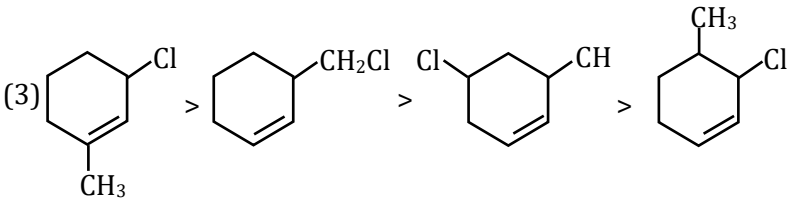
For More Join: @IITJEE_Advanced

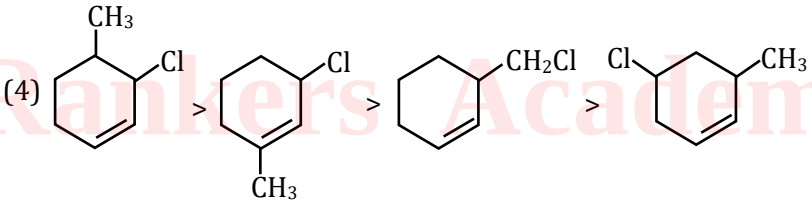
31. The **correct** order of reactivity of the given chlorides with acetate in acetic acid is :

[JEE (Main) 2021]

(1) 

(2) 

(3) 

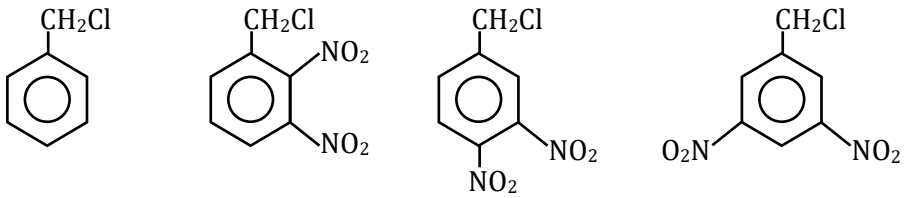
(4) 

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CRM116

32. The decreasing order of reactivity of the following compounds towards nucleophilic substitution (S_N2) is :

[JEE (Main) 2020]



(I) (II) (III) (IV)

(1) (IV) > (II) > (III) > (I)

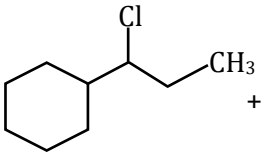
(2) (II) > (III) > (IV) > (I)

(3) (II) > (III) > (I) > (IV)

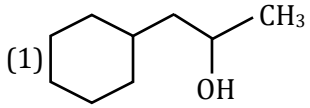
(4) (III) > (II) > (IV) > (I)

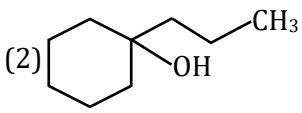
CRM117

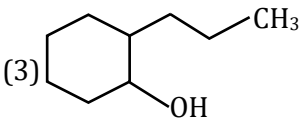
For More Join: @IITJEE_Advanced

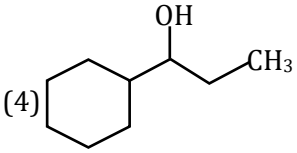
33.  + NaOH $\xrightarrow{H_2O}$ Major Product "A"

Consider the above chemical reaction. Product "A" is: **[JEE (Main) 2024]**

(1) 

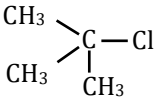
(2) 

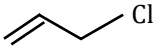
(3) 

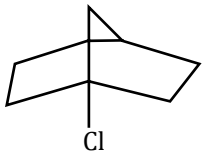
(4) 

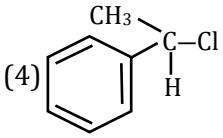
CRM118

34. Which one of the following compounds is inactive towards S_N1 reaction? **[JEE (Main) 2022]**

(1) 

(2) 

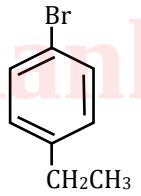
(3) 

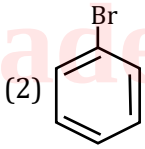
(4) 

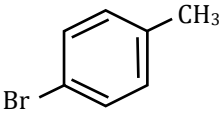
CRM119

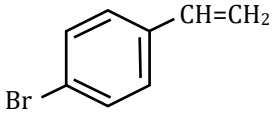
35. A $\xrightarrow[(iii)\,H_2O/H^+]{(i)\,Cl_2,\Delta\ (ii)\,CN^-}$ 4-Bromophenyl acetic acid. **[JEE (Main) 2022]**

In the above reaction 'A' is

(1) 

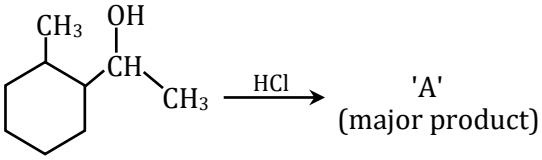
(2) 

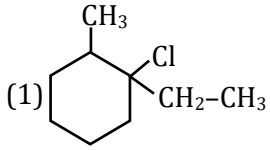
(3) 

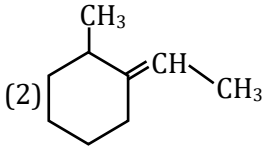
(4) 

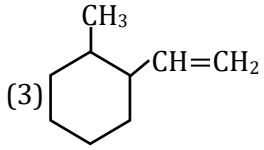
CRM120

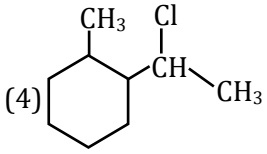
36. What is the final product (major) 'A' in the given reaction ? **[JEE (Main) 2021]**



(1) 

(2) 

(3) 

(4) 

CRM121

Rankers Academy JEE

For More Join: @IITJEE_Advanced

37. In the given reaction 3-Bromo-2, 2-dimethyl butane $\xrightarrow{C_2H_5OH}$ 'A' (Major Product) Product A is:

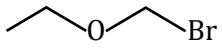
[JEE (Main) 2021]

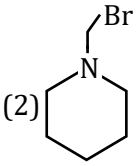
(1) 2-Ethoxy-3, 3-dimethyl butane
(2) 1-Ethoxy-3, 3-dimethyl butane
(3) 2-Ethoxy-2, 3-dimethyl butane
(4) 2-Hydroxy-3, 3-dimethyl butane

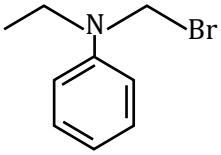
CRM122

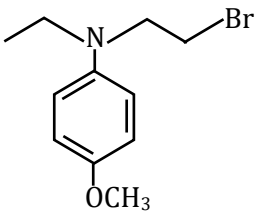
38. Which of the following compounds will form the precipitate with aq. $AgNO_3$ solution most readily ?

[JEE (Main) 2020]

(1) 

(2) 

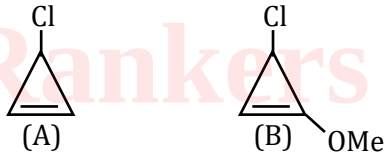
(3) 

(4) 

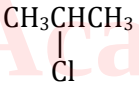
CRM123

39. The decreasing order of reactivity of the following organic molecules towards $AgNO_3$ solution is :

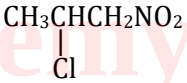
[JEE (Main) 2020]



(A) (B)



(C)



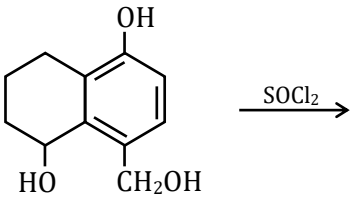
(D)

(1) (A) > (B) > (D) > (C)
(2) (A) > (B) > (C) > (D)
(3) (C) > (D) > (A) > (B)
(4) (B) > (A) > (C) > (D)

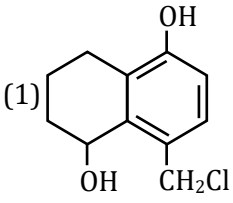
CRM124

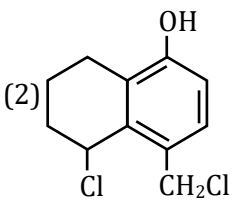
40. Identify A in the given reaction.

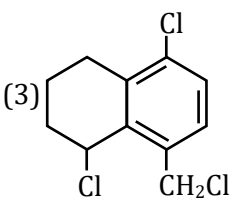
[JEE (Main) 2021]

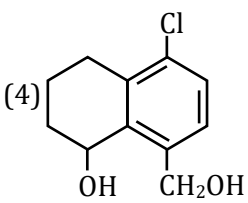


$\xrightarrow{SOCl_2}$ A (Major Product)

(1) 

(2) 

(3) 

(4) 

CRM125

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Identify the set of reagents / reaction conditions 'X' and 'Y' in the following set of transformation: **[IIT 2002]**

$CH_3 - CH_2 - CH_2Br \xrightarrow{X}$ Product $\xrightarrow{Y}$ $CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3$

(A) X = dilute aqueous NaOH, 20°C; Y = HBr / acetic acid, 20°C

(B) X = concentrated alcoholic NaOH, 80°C; Y = HBr/ acetic acid 20°C

(C) X = dilute aqueous NaOH, 20°C; Y = Br_2 / $CHCl_3$, 0°C

(D) X = concentrated alcoholic NaOH, 80°C; Y = $Br_2/CHCl_3$, 0°C

CRM126

2. An S_N2 reaction at an asymmetric carbon of a compound always gives. **[IIT 2001]**

(A) an enantiomer of the substance

(B) a product with opposite optical rotation

(C) a mixture of diastereomers

(D) a single stereoisomer

CRM127

3. The compound that will react most readily with NaOH to form methanol is **[IIT 2001]**

(A) $(CH_3)_4N^{\oplus}I^{\ominus}$ (B) CH_3OCH_3 (C) $(CH_3)_3S^{\oplus}I^{\ominus}$ (D) $(CH_3)_3CCl$

Rankers Academy JEE

CRM128

4. The order of reactivity of the following alkyl halides for a S_N2 reaction is: **[IIT 2000]**

(A) R–F > R–Cl > R–Br > R–I (B) R–F > R–Br > R–Cl > R–I

(C) R–Cl > R–Br > R–F > R–I (D) R–I > R–Br > R–Cl > R–F

CRM129

5. Which of the following has the highest nucleophilicity? **[IIT 2000]**

(A) $F^{\ominus}$ (B) $OH^{\ominus}$ (C) $CH_3^{\ominus}$ (D) $NH_2^{\ominus}$

CRM130

6. The products of reaction of alcoholic $AgNO_2$ with ethyl bromide are **[IIT 1991]**

(A) Ethane (B) Ethyl nitrite (C) Nitroethane (D) Ethyl alcohol

CRM131

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	A	A	B	A	B	C	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	D	A	C	D	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	D	B	D	C	A	A	C	D	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	B	C	A	A	B	D	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	B	C	C	C	B	B	A	B	C
Que.	51	52	53	54	55	56				
Ans.	A	C	D	D	C	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	B,C,D	B,D	A,B	A,B,D	B,D	A,B	A,B,C	C,D	C,D
Que.	11									
Ans.	A,B,C,D									

Rankers Academy JEE

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	5	3	2	4	4	6	4	12	3
Que.	11	12	13	14	15	16	17	18		
Ans.	5	6	5	3.00	4.00	4.00	3.00	8.00		

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	3	2	2	4	4	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	3	1	4	4	2	2	2	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	3	3	1	3	4	1	2	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	2	2	3	3	1	3	2	4	2

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6
Ans.	B	D	A	D	C	C

For More Join: @IITJEE_Advanced